When the Model Screens Off the Harness: Per-Decision Cross-Layer Causal Credit in LLM Agents

[TODO: autores]

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Abstract

LLM agents interleave decisions from two layers: the *model* (tool calls) and the *harness* (context management, retries, termination). We show, via deterministic counterfactual replay with a *measured* noise floor (exact fidelity under explicit serving premises), that per-decision causal contributions $C(H)$, $C(M)$ and their interaction $I(H, M) = C(H, M) - C(H) - C(M)$ are identifiable on the same trajectory. Interaction is *regime-dependent*: with budget slack, the model action screens off the harness decision ($C_{HM} = C_M$, 0 breaks in 57 points across three configurations); under budget pressure, screening-off breaks (8/65 across three pressure levels; $p = 0.005$ point-level, $p = 0.125$ task-clustered — we report both) at *recurring* decision points, with a non-monotonic dependence on pressure intensity partially mediated by reward saturation. The noise floor and the slack regime replicate on a second model (Qwen3-1.7B); the pressure regime is established on one model — the second model's null is itself explained by saturation (1 non-saturated point). This implies that single-layer credit double-counts shielded decisions. We then train the harness layer with interaction-aware credit against outcome-only, single-layer, and untrained controls (dose-matched in total rollouts, 3 seeds, twice — the second run with a pre-registered optimization fix): no credit signal learns the threshold policy that is expressible by the policy class and dominant on held-out tasks (held-out R_{eff} 0.398 vs. 0.237 for the best fixed policy), and credit-driven arms add collapse risk that the outcome-only arm avoids. A pre-registered audit (60 points, two-greedy-replay recomputation, 58/60 sign agreement) rules out the estimator as the cause: the binding constraint is *exploration*, not credit quality. We report this two-stage identified negative result as a design lesson for cross-layer training benchmarks.

1 Introduction

An LLM agent is not a model. It is a model wrapped in a *harness*: a program that manages context, decides when to summarize or truncate history, retries failed calls, and terminates episodes. In deployed agents the harness makes as many decisions per episode as the model does, and those decisions — especially context management — destroy or preserve the information the model conditions on. Yet the dominant training paradigm treats the trajectory as a monolith: outcome-level RL (e.g., GRPO-style policy gradients) assigns one scalar return to every decision, regardless of which layer made it, and recent per-step credit methods attribute within a single layer only. If a harness decision was harmless *because* the model's next action compensated for it, single-layer credit will bill the harness for an effect the model screened off — a double-counting error that is invisible unless both layers are measured on the same trajectory.

This paper asks the question directly: on the same trajectory, what is the causal contribution of a harness decision, of a model decision, and of their interaction? We answer it with deterministic counterfactual replay. Given a recorded trajectory, we re-execute it from any decision point with exactly one decision changed and the entire prefix forced, and define per-decision credit as $C(d) = R_{\text{orig}} - R_{\text{cf}}$. Replaying pairs and joint flips yields $C(H)$, $C(M)$, $C(H, M)$ and the interaction $I(H, M) = C(H, M) - C(H) - C(M)$ at a single decision pair. Because our replay is *exactly* deterministic (a measured, not assumed, property — and one that fails silently if vLLM's automatic prefix caching is left on), every nonzero credit is signal.

The headline finding is that the interaction term is *regime-dependent*. When the agent has budget slack, the model action screens off the harness decision exactly: $C_{HM} = C_M$ in 57 out of 57 measured points across three configurations. Under budget pressure the shield cracks — 8 breaks in 65 points, concentrated at recurring decision points — and the dependence on pressure intensity is non-monotonic, partially mediated by reward saturation: interaction is only detectable inside a *competence window* where the agent neither aces nor fails everything. The practical corollary is negative for naive pipelines: in the slack regime, single-layer harness credit systematically double-counts effects the model absorbs.

We then test whether interaction-aware credit improves *training* of the harness layer, with pre-registered predictions, dose-matched controls, and three seeds per arm — twice. The honest outcome is a two-stage identified negative: optimization instability first, and, after a pre-registered fix, stable training that still fails to discover a threshold policy which is expressible by the policy class and dominant on held-out tasks. A pre-registered audit exonerates the credit estimator; the binding constraint is exploration. We report this as a design lesson for cross-layer training benchmarks, not as a method contribution.

Contributions. (1) Deterministic replay infrastructure with measured noise floor *under explicit identification premises* (fixed config, sequential execution, prefix caching off) — including a cautionary finding: vLLM automatic prefix caching silently breaks greedy-decoding determinism across differing histories; (2) per-decision counterfactual decomposition $C(H)$, $C(M)$, $I(H, M)$ on the same trajectory; (3) anatomy of screening-off: a structural component (the sampled a' re-injects destroyed information) proven as a property of the estimand, and an empirical component (the live downstream harness re-triggers the intervened decision); (4) control tasks showing the method distinguishes harmless from harmful harness decisions by construction — including one audited exception where a harness decision had a *behavioral* (not informational) effect; (5) a pre-registered training comparison — interaction-aware credit vs. outcome-only, single-layer credit (a reproduction of the CHILL-Harness principle in our environment), and a zero-credit masking control (which, under our greedy tie-break at $\theta = 0$, is de facto an untrained keep-always policy — we state this explicitly), dose-matched in total rollouts — whose honest outcome is a *two-stage identified negative*: (act 1) with raw features and high learning rate, every trained arm collapses seed-dependently into one of two policy attractors; (act 2) with a pre-registered optimization fix (centered features, lower learning rate, gradient clipping), training is stable and moves in the right direction, yet no arm discovers the threshold policy that provably exists on held-out tasks — and credit-driven arms retain collapse risk (1/3 seeds) that outcome-only avoids (0/3). A pre-registered audit (58/60 sign agreement between logged credits and two-greedy-replay recomputation) rules out the estimator: exploration, not credit quality, is the binding constraint.

2 Related Work

Counterfactual replay for agent credit is an active area, but every existing system we are aware of attributes within a single layer. CHILL-Harness [3] intervenes on harness decisions and allocates replay budget adaptively, but never measures the model layer or the interaction; our single-layer training arm (§6) is an explicit reproduction of its principle in our environment, which turns the comparison into a result. CAR [2] formalizes per-step do-operations with Shapley attribution under a replay budget, within one decision stream. C3 [1] audits credit fidelity against exact replay ground truth (the same validation metric we adopt) but decomposes across *agents*, not across the model/harness boundary inside one agent. A recent negative result [7] shows that popular step-level credit signals fail to beat chance against replay ground truth; it sets the evidential bar our critic section inherits (dose-matching, pre-registration, heuristic baselines). On the training side, Co-Harness [4], HarnessCompass [5] and HASE [6] jointly optimize harness and weights, but without a per-decision causal signal — they optimize the composition blindly. Agent Lightning [8] provides rollout infrastructure for agent RL without addressing attribution. Table 1 summarizes: no prior work measures $C(H)$, $C(M)$ and $I(H, M)$ on the same trajectory, which is precisely where the double-counting error lives.

Method	counterfactual via replay	measures $C(H)$	measures $C(M)$	$I(H, M)$ on same trajectory	trains with I
CHILL-Harness	✓	✓	–	–	–
CAR	✓	–	✓	–	–
C3 v2	✓	(por agente)		–	–
HASE	–	✓	–	–	–
Ours	✓	✓	✓	✓	tested†

Table 1: T1 — positioning. †Tested with dose-matched controls and pre-registration; the outcome is an identified negative (§6). [TODO: verify cells against primary sources before submission]

3 Setup: A Two-Layer Agent with Deterministic Replay

3.1 Agent, decision schema and recorder

The agent is deliberately minimal: a coding agent in a sandboxed environment that reads a task description, writes files, and runs tests. Two layers make decisions. The **model layer** (Qwen3-4B served by vLLM, greedy decoding) chooses tool calls: `write_file`, `run_tests`, `read_file`, `finish`. The **harness layer** is a scripted policy that at every turn makes a `context_policy` decision (`keep_context` vs. `summarize_context`, the latter compressing history to a fixed-length prefix) and a `termination` decision. Every decision — model or harness — is recorded with a typed schema: decision id, layer, state before, available actions, chosen action, observation, token costs, and parent/child links. Reward is deterministic: the fraction of hidden pytest tests passed, so $R \in [0, 1]$ with resolution set by the number of tests. Cost-adjusted reward is $R_{\text{eff}} = R - \lambda \cdot \text{prompt tokens}/10^5$ with $\lambda = 25$, fixed analytically before any training and never recalibrated.

3.2 Replay and interventions

Replay re-executes a recorded trajectory through the *live* agent loop with a forced-action queue: recorded decisions are matched against the loop's requests (peek-match on decision type and state hash) and forced, up to the intervention point; from there the loop runs free. An intervention replaces exactly one decision (or, for $C(H, M)$, one harness–model pair). The null intervention — force everything, change nothing — must reproduce R exactly; §4.1 shows it does, and what serving configuration is required for it to. All replays run sequentially against a single vLLM instance with a fixed config.

3.3 Tasks

We use 30 synthetic coding tasks in four designed strata, plus 10 held-out tasks. **V2** and **L** tasks place critical constants beyond the first 240 characters of history, so a summarize decision is *informationally destructive* at known positions; L tasks are longer variants. **S** tasks are built so that success requires harness and model decisions to cooperate (synergy by design). **C** tasks are controls: all information needed survives any summary, so the construct predicts $C(H) = 0$. Reward saturation (episodes at $R \in \{0, 1\}$) is endogenous to the number of tests per task; we treat it as a design choice and analyze its consequences explicitly (§4.3, §7).

4 Measuring Cross-Layer Credit

4.1 Noise floor

Every causal claim below is only as good as the replay. We therefore *measure* the noise floor rather than assume it: the pre-registered floor for each configuration is $\max |\Delta R|$ over null replays (replay with no intervention). Across seven configurations of Qwen3-4B (1,230 null replays) and two configurations of Qwen3-1.7B, **every null replay reproduced R exactly**: the floor is 0.0 in all nine configurations (Figure 1a). This is a measurement under explicit premises — fixed serving config, strictly sequential requests, prefix caching off — not a guarantee.

The premises matter. vLLM V1 enables automatic prefix caching (APC) by default, and with APC on, prefill numerics depend on the KV-cache state left by *previous* requests: greedy decoding diverges across identical prompts with different serving histories, even when requests are sequential. In our APC-contaminated run, 7 of 270 null replays were inexact and the floor rose to 0.417 (Figure 1b) — large enough to swamp most true credits. Sequential execution alone is *not* sufficient; APC must be disabled. We flag this because any replay-based attribution system built on default vLLM settings inherits this silent failure.

4.2 $C(H)$ and $C(M)$

With an exact floor, per-decision credits are directly interpretable: any $C \neq 0$ is a real causal effect of that single decision. Figure 2 shows the distributions over 248 harness points and 66 model points, by stratum. Three regularities matter downstream. First, direction is everything: flipping `keep`→`summarize` at an information-critical point yields large positive $C(H)$ (the decision to keep was worth up to +0.89), while the reverse direction is near zero — we therefore never aggregate across directions. Second, credits are sparse: most decisions are causally inert, and mass concentrates at information-destruction points predicted by task design (V2/L strata). Third, $C(M)$ at matched points is frequently nonzero when $C(H)$ is not, foreshadowing the screening-off structure of

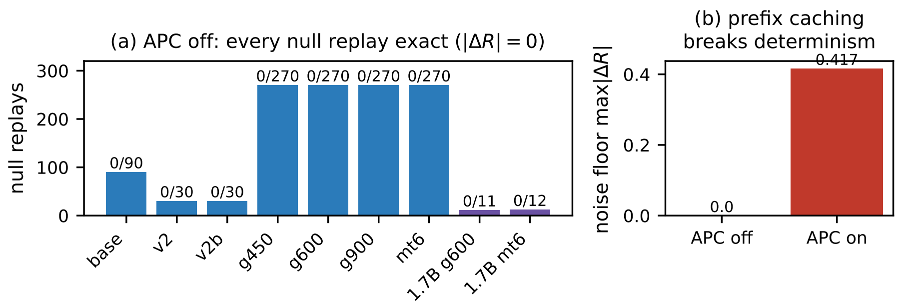

Figure 1: **F1 — Measured noise floor.** (a) Null replays per configuration (Qwen3-4B blue, Qwen3-1.7B purple); every one of ~1,250 null replays is exact, floor = 0.0. (b) With vLLM automatic prefix caching on, the floor rises to 0.417 (7/270 inexact): greedy determinism breaks silently.

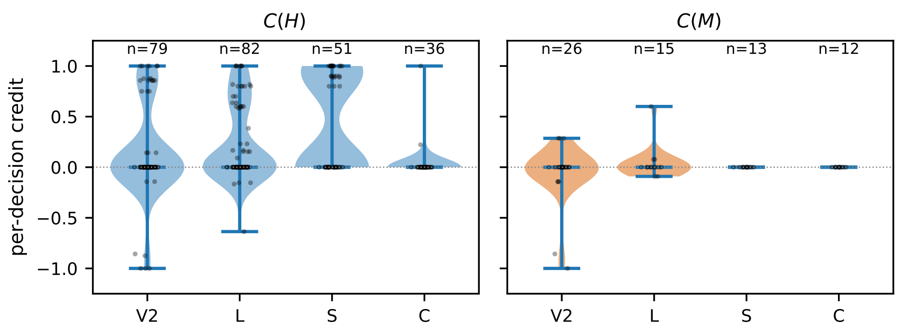

Figure 2: **F2 — Per-decision credit by layer and stratum.** $C(H)$ (248 points) and $C(M)$ (66 points) against an exact zero floor. Strata: V2/L information-destruction, S designed synergy, C controls.

§4.3.

4.3 Interaction $I(H, M)$ and its anatomy

The central measurement flips a harness decision and a model decision at the same point, alone and jointly. **With budget slack** (max 12 turns; three budget configurations g450/g600/g900), the model action screens off the harness decision *exactly*: $C_{HM} = C_M$ in **57/57** points — the sampled alternative model action re-establishes what the harness destroyed, so the harness flip adds nothing beyond the model flip. **Under budget pressure** (max 4/6/8 turns), screening-off breaks in **8/65** points (slack vs. pressure: $p = 0.005$ point-level hypergeometric; $p = 0.125$ task-clustered sign test over 3 breaking tasks — we report both, and note the point-level test treats points within a task as independent).

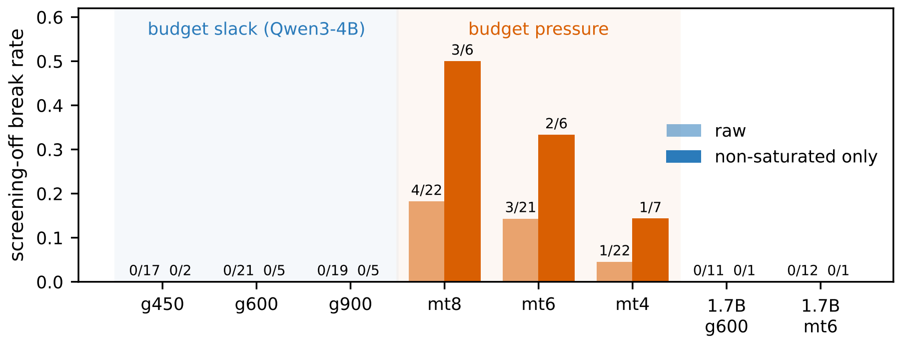

Figure 3: **F3 — Screening-off is regime-dependent.** Break rate of $C_{HM} = C_M$ per configuration, raw and conditioned on non-saturated points. Slack: 0/57 raw (0/12 non-saturated). Pressure: 8/65 raw (6/19 non-saturated, $p = 0.037$). Qwen3-1.7B (purple) replicates slack; its pressure null is explained by saturation (only 1 non-saturated point).

Breaks recur at the *same* decision points across configurations (`l_log_parser` idx 8 in 3/3 pressure configs; `api_router` idx 1 and `l_vending_machine` idx 11 in 2 each), which is what a real mechanism, rather than noise, predicts.

The dependence on pressure intensity is *non-monotonic*: break rates are 0/21 (mt12), 4/22 (mt8), 3/21 (mt6), 1/22 (mt4) — an inverted U. Our pre-registered monotonicity test failed (Cochran–Armitage $p = 0.38$); we report that failure and analyze the cause. The mediator is **reward saturation**: at mt4 the agent fails most tasks outright (R pinned at 0), so no flip can make things worse and interaction is masked. Conditioning on non-saturated points sharpens the contrast (slack 0/12 vs. pressure 6/19, $p = 0.037$) and partially restores the mt4 rate ($0.045 \to 0.143$), though the ordering mt8 $>$ mt6 $>$ mt4 persists — saturation mediates the window partially, not fully (Figure 3). We call the resulting picture a *competence window*: cross-layer interaction is detectable only where the agent neither saturates success nor failure.

Anatomy of the shield. Why does screening-off hold so often? A sign-test of the mechanism's core prediction — that the model flip weakly dominates the harness flip at shielded points — holds in 75/78 points ($P = 2.6 \times 10^{-19}$, pre-registered). Decomposing the shielded points for which we have mechanism annotations: in 23 the sampled alternative action a' *structurally re-injects* the destroyed information (a property of the estimand: a' is sampled from the model conditioned on the un-summarized state, so it can carry the information across the cut); in 22 the *live downstream harness* re-triggers summarization anyway, equalizing the branches; 9 show both, 1 neither. Screening-off is thus partly a theorem about the estimator and partly an empirical fact about closed-loop harness behavior — and we found genuine non-saturated synergy where it breaks (`l_vending_machine`: $C_H = C_M = -0.09$, $C_{HM} = 0$, so $I = +0.18$: either flip alone hurts, together they cancel).

Consequence for credit assignment. At every shielded point, single-layer harness credit $C(H)$ bills the harness for an effect the model absorbs: interaction-corrected credit $C_{HM} - C_M$ is zero while $C(H)$ is not. In the slack regime this is not an edge case — it is *all* measured points.

4.4 Control tasks and the `c_temp_label` exception

On control tasks, where summaries destroy nothing by construction, the method returns $C(H) = 0$ in 11/12 measured points — the construct behaves. The exception is informative: `c_temp_label` showed $C = +0.22$ at turn 0 despite all task-critical information surviving the summary. Audit of the replay pair shows the effect is *behavioral*, not informational: the summarized context changes the model's subsequent formatting behavior, which a strict test then penalizes. $C(H)$ measures total causal effect through *any* pathway; the controls certify the informational pathway is clean, and the exception demonstrates the behavioral pathway exists and is detectable.

5 The Structure of Harness Credit

Can $C(H)$ be *predicted* from cheap features, so that replay is needed only to calibrate? We trained linear and GBM critics on 248 ground-truth points (features: turn, context size, position, test progress; task-clustered evaluation) against the baselines mandated by [7]: position and context-size heuristics. The honest answer is that **the critic never beats the heuristics** (Figure 4): pooled ranking 0.718 (GBM) vs. 0.796 (|context|); per stratum, L is a tie (0.747 vs. 0.741), V2 the heuristic wins (0.693 vs. 0.796), and S saturates for everything (AUROC 1.0). For detection ($|C| > 0$) the GBM has a modest edge (AUROC 0.846 vs. 0.735/0.785) with overlapping CIs.

We report this not as a failed component but as a finding about the signal: **harness credit is mostly a function of context size and position**. That is exactly what the information-destruction mechanism predicts — how much is destroyed, and how late — and it means practitioners can get most of the ranking value of $C(H)$ from features they already log, reserving replay for calibration and for the interaction term, which no cheap feature predicts.

6 Training with Cross-Layer Credit

Does better credit train a better harness? We train the harness `context_policy` (a logistic policy over 5 features: bias, turn fraction, context tokens, test progress, message count) with four arms, **dose-matched in total rollouts** (episodes + credit replays): (1) *outcome-only* policy gradient; (2) *single-layer* $C(H)$ credit

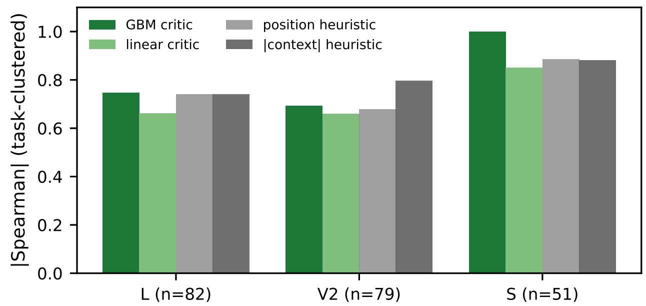

Figure 4: **F4 — Critics vs. trivial heuristics per stratum** (task-clustered |Spearman|). The critic never beats the best heuristic; $C(H)$ is structured by position and context size.

— an explicit reproduction of the CHILL-Harness principle in our environment; (3) *interaction-corrected* credit $C_{HM} - C_M$; (4) a *zero-credit* masking control (credit $\equiv 0$, no replays — which under our greedy tie-break at $\theta = 0$ is de facto an untrained keep-always policy; we state this explicitly). The pre-registered success criterion (GATE 3) required arm 3 to beat both arm 2 and arm 4. Calibration established the target is learnable in principle: on held-out tasks a simple context-size threshold policy (thr600) achieves $R_{\text{eff}} = 0.398$, dominating keep-always (0.237) and summarize-always (-0.464); thr600 is expressible by the policy class.

Act 1: seed-dependent collapse. With raw features and lr $= 0.5$, every trained arm collapsed into one of the two fixed policies, with the attractor determined by seed (outcome: k/k/s; $C(H)$: s/s/k; $C_{HM}-C_M$: k/s/s). GATE 3 failed and we report it. Two hypotheses remained: the credit estimator is broken, or the optimization is. A pre-registered audit recomputed 60 logged credits by fresh two-greedy-replay: **58/60 sign agreement, zero sign flips toward harmful** (mean discrepancy -0.0015). The estimator is exonerated.

Act 2: stable optimization, no discovery. With the pre-registered fix (feature centering, lr $= 0.1$, gradient clipping), training is sane: gradients bounded, learned weights point the right way (negative bias, positive weight on context tokens, implicit decision thresholds at 2.5k–5.8k tokens). Yet **no arm in any seed discovers the threshold policy**: all non-collapsed runs converge to keep-always (held-out $R_{\text{eff}} = 0.237$ vs. thr600's 0.398; Figure 5). Worse, credit-driven arms retain collapse risk the outcome-only arm avoids ($C(H)$: 1/3 seeds; $C_{HM} - C_M$: 1/3; outcome-only and zero: 0/3).

Diagnosis. The learned thresholds sit at 2.5k–5.8k tokens, but under keep-heavy behavior the training distribution rarely visits such states: the informative

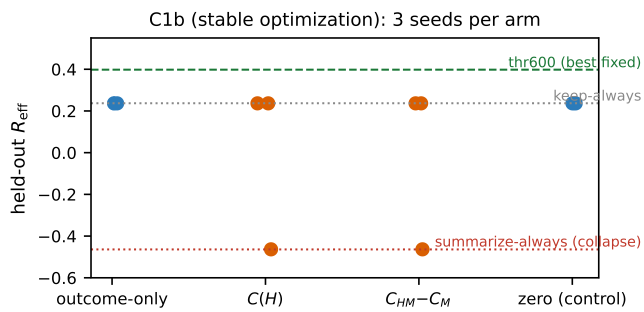

Figure 5: **F5 — Training outcome (C1b, stable optimization).** Held-out R_{eff}, 3 seeds per arm. No arm reaches the expressible thr600 policy; credit-driven arms (orange) carry collapse risk that outcome-only avoids.

region is barely explored, so the gradient toward thr600 is starved regardless of how accurate the per-decision credit is. **The binding constraint is exploration, not credit quality.** We report this two-stage identified negative as a design lesson: a cross-layer training benchmark must guarantee coverage of the states where layers interact, or credit fidelity is moot.

7 Threats to Validity and Discussion

Synthetic tasks and external validity. Our tasks are designed, not sampled from a public benchmark, and this is a real limitation — but a motivated one. The measurement requires (i) control over the *position* of destroyable information, to separate informational from behavioral effects; (ii) per-stratum calibrated recoverability, which underlies the control tasks; and (iii) deterministic fractional reward, which underlies the exact floor. No public benchmark provides these jointly. The exact missing experiment is a multi-turn MBPP+ adaptation (∼100 tasks, 1–2 GPU-days); we declare it rather than claim it. We position this as a methods paper: the transferable artifact is the recorder/replay/forced-queue infrastructure, the APC finding, and the measurement protocol.

One model powers the pressure regime. The floor and slack regime replicate on Qwen3-1.7B; the pressure-regime break rate is established on Qwen3-4B only. The 1.7B null (0/12) is itself explained by the mediator we identified: 11/12 of its pressure points are reward-saturated, leaving one informative point — zero power (under the 4B break rate, $P(0 \text{ breaks}) \approx 0.21$). This is the competence window operating as predicted, but a second *powered* model is needed to call the pressure regime general.

Clustered inference. Points within a task are not independent. At the task level the pressure-vs-slack contrast is 3 tasks vs. 0 (sign test $p = 0.125$); at the point level $p = 0.005$. We report both and lean on the mechanism-level sign test (75/78, $P = 2.6 \times 10^{-19}$) and cross-configuration recurrence of break points as the stronger evidence.

Estimator coupling. $C(M)$ and I share the sampled alternative action a': they are not independent evidence, and the structural component of screening-off (§4.3) is partly a property of the estimand — a' sampled from the unsummarized state can re-inject information. We quantified this (23/52 shielded points) rather than hiding it; an estimand with a' sampled from the summarized state would measure a different, also legitimate, quantity.

Saturation is endogenous. Reward resolution is set by test count, so saturation — our mediator — is partly a design artifact. The competence-window analysis conditions on it explicitly; the inverted-U at mt4 is only partially explained (conditioned rate $0.045 \rightarrow 0.143$; the ordering mt8 > mt6 > mt4 persists), so the window remains a partially supported hypothesis, discussed but not claimed as a contribution.

Serving premises. The exact floor holds under fixed config, sequential requests, and APC off. It is a measured property of this stack, not a theorem; any deviation (batching, caching, config drift) voids it and must be re-measured.

8 Conclusion

We built the first apparatus that measures, on the same trajectory, the causal credit of harness decisions, model decisions, and their interaction, against an exactly-zero measured noise floor. The measurement yields a crisp, regime-dependent picture: with budget slack the model screens off the harness completely (0/57), so single-layer harness credit double-counts; under budget pressure the shield breaks at recurring points (8/65), inside a competence window shaped by reward saturation. Better credit did not, however, train a better harness: a two-stage, pre-registered negative — with the estimator exonerated by audit — locates the binding constraint in exploration, not attribution. We offer the infrastructure, the regime-dependence finding, the APC determinism warning, and a pre-registration ledger of 14 predictions (including 3 failures) as the contributions, and the training negative as the design brief for the benchmark this field still needs.

A Pre-Registration Ledger

All confirmatory analyses were pre-registered in the project's execution plan before the corresponding data were collected. Table 2 lists all 14 pre-registrations with their outcomes, including the three that failed. Failed predictions are reported in the main text where they occur.

#	Pre-registered prediction / rule	Outcome	Status
1	Noise floor per config = max $\lvert\Delta R\rvert$ over nulls	0.0 in 7+2 configs, 2 models	held
2	Phase-A yield / stopping rule	followed as written	held
3	Critic evaluation statistic (task-clustered Spearman)	mixed result reported as-is	held
4	Pre/post feature split for critic	followed	held
5	Structured a' = distinct estimand, reported separately	followed	held
6	C1 arms dose-matched in total rollouts	followed	held
7	Credit directions never aggregated	followed	held
8	Saturation analyses outside confirmatory set	followed	held
9	GATE 3: arm 3 must beat arms 2 *and* 4	not met	**failed**
10	Screening-off mechanism sign test	75/78, $P = 2.6 \times 10^{-19}$	held
11	Estimator audit: sign agreement on 60 recomputed credits	58/60, 0 harmful flips	held
12	C1b primary outcome: held-out $R_{\text{eff}} > 0.30$, $\geq$2/3 seeds	no arm reached it	**failed**
13	Dose-response: monotonic break rate in pressure	inverted U ($p = 0.38$)	**failed**
14	D2c replication: floor, slack, pressure on Qwen3-1.7B	floor+slack replicate; pressure inconclusive	partial

Table 2: Pre-registration ledger. Failures are results, not omissions: #9 and #12 constitute the two-stage training negative (§6); #13 led to the competence-window analysis (§4.3).

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